This document is a summary of chapters 1 and parts of 2 of winter's biomechanics book 2022 Edition.
As a general note this summary is from my perspective as a robotics engineer learning about the basics of biomechanics.

Table of Contents

Chapter 1 : BIOMECHANICS AS AN INTERDISCIPLINE
Chapter 2 : SIGNAL PROCESSING
  2.0 intro
  2.1 Auto- and cross- correlation analysis
  2.2 Frequency analysis:
    2.2.2 Discrete fourier (harmonic) analysis
    2.2.3 Fast fourier transform (FFT):
    2.2.4 Applications of Spectrum analysis

Chapter 1 : BIOMECHANICS AS AN INTERDISCIPLINE

This chapter introduces the following definitions.

- Kinesiology : The broad science of human movement combining physiology, psychology, nutrition, exercise , biomechanics, sports, motor development and learning to understand how humans move and function.
- Biomechanics : The study of human movement. Measure, observe, analyze, assess and interpret.
- Analyze :Using math on measured data to produce variables which were not directly measurable. Ex use of a low pass filter.
- Assessment: Using the information produced from analysis to make decisions assess the results.
- Interpretation: Understand why the movement works the way it is.
- Pathology/ Pathological population : Pathology is a term used to describe different movement or abnormal behaviour. Pathological population in the context of the book is used for people with diseases that affect their movement. Like a stroke, injury etc.

Chapter 2 : SIGNAL PROCESSING

2.0 Introduction

All biomechanics signals are time varying whether they are EMG signals, or force & acceleration signals coming from transducers, they can benefit from signal processing.
Important definitions
Harmonics : Harmonic is a basic frequency of a cyclical action. A stride, a gait cycle, etc. For example a stride that repeats every second has a first harmonic of 1 Hz and other harmonics are just a multiple of the first.

2.1 Auto- and cross- correlation analysis

Auto-correlation : is a technique used to find similarities “correlations” within the signal itself comparing values of the signal now to its values in the past, present and future.

Cross-correlation : analysis used to evaluate how a signal is correlated with another signal. Over past,present and future points in time.

The correlation coefficient is between -1 and 1.
Perturbation is simply an interruption , in this context an external one. Ex a change of terrain or change of external environment while walking, etc. In many cases it can be introduced for the purpose of assessment or other purposes.

Applications of auto-correlation : autocorrelation asks does the signal repeat and how often? Is there a hidden pattern / noise?

1. Detects hum (a low frequency interference) Ex, a 50/60Hz power line.
2. Find periodicity
3. Detect hidden patterns
4. Find dominant frequency content.

Applications of cross-correlations : is signal A related with signal B and does it lead or lag ?

1. Measure timing relationships between joints or muscles. *Ex compare EMG of quads vs hamstring to see which activates first.*
2. Compare left vs right limb signals. Useful in gait symmetry analysis.
3. Determine coordination between segments. *Ex: How ankle angle is related to knee angles in walking.*
4. Identify delays in neuromuscular responses. Ex how long after a perturbation does a muscle/EMG signal respond.
5. Compare two measurement systems. For example how well an accelerometer signal matches a motion capture velocity signal.

2.2 Frequency analysis:

Time domain : Simply the evolution of a signal over time , measurement of a volt EMG signal , gait cycle movement etc.

The frequency domain reveals information the time domain doesn't. Simply put the frequency domain plots all the harmonics of the signal. On the horizontal x-axis plots the frequencies in the signal and on the vertical y-axis their corresponding power or amplitude.

Fourier transform : Fourier discovered that any signal can be made up of sins/cosins summed up together. With the same concept any signal can be broken down to sins + cosines and we can find out the frequency content of that signal.

The knowledge of the frequency content of any signal is mandatory before making decisions such as sampling rate for A/D converters and in general it reveals many information about the signal.(like revealing noise , dominant frequencies , highest frequency)

2.2.2 Discrete fourier (harmonic) analysis

1. Alternating signals. (AC for alternating current) a signal that continuously changes over time. Which may be periodic (some what a gait cycle) or completely random (EMG) or a combination. AC might have a DC component which is called DC bias which about the AC fluctuates.
2. Frequency content : As said before any wave can be made up of a sum of sines and cosines.

down into its harmonic components. The sum of the proper amplitudes of these harmonics is called a *Fourier series*.

Thus, a given signal $V(t)$ can be expressed as:

$$V(t) = V_{dc} + V_1 \sin(\omega_0 t + \theta_1) + V_2 \sin(2\omega_0 t + \theta_2) + \cdots + V_n \sin(n\omega_0 t + \theta_n) \tag{2.10}$$

where $\omega_0 = 2\pi f_0$, and θ_n is the phase angle of the nth harmonic.

For example, a square wave of amplitude V can be described by the Fourier series of odd harmonics:

$$V(t) = \frac{4V}{\pi}\left(\sin\omega_0 t + \frac{1}{3}\sin 3\,\omega_0 t + \frac{1}{5}\sin 5\,\omega_0 t + \cdots\right) \tag{2.11}$$

A triangular wave of duration $2t$ and repeating itself every T seconds is:

$$V(t) = \frac{2Vt}{T}\left[\frac{1}{2} + \left(\frac{2}{\pi}\right)^2 \cos\omega_0 t + \left(\frac{2}{3\pi}\right)^2 \cos 3\omega_0 t + \cdots\right] \tag{2.12}$$

The re-construction of a measured signal x(t) can be done by the following using fourier:

$$b_n = \frac{2}{T}\int_0^T x(t)\sin n\omega_0 t\,dt \tag{2.14}$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1}\left(\frac{a_n}{b_n}\right) \tag{2.15}$$

Digital equivalent:

$$a_n = \frac{2}{N}\sum_{i=0}^{N} x_i \cos(n\omega_0 i/N) \tag{2.16}$$

$$b_n = \frac{2}{N}\sum_{i=0}^{N} x_i \sin(n\omega_0 i/N) \tag{2.17}$$

For each of the harmonics n, N calculations are needed. For n = N/2 harmonics there are N^2/2 for sine and cosine coefficients. The total number of calculations is N^2.

2.2.3 Fast fourier transform (FFT):

FFT became necessary because of the large number of calculations required by the DFT. The idea is that they want to save calculation time.

How does FFT save calculation time?

Example: For the first harmonic n=1 we must calculate N sine and N cosine values. And for the second harmonic we re-calculate every sine and cosine value, and so on. To save time the FFT calculates the sine and cosine values for the first harmonic and this forms a “look-up” table up to the highest harmonic. Because Cosine and sine values repeat in a predictable pattern and once there is a pattern established the look up table is set the higher

order harmonic is found in the table. The other way FFT saves time is by grouping same values of sine and cosine to be multiplied to their corresponding x(i) value. For example sine values of 0.707 are a group, these values will multiply x2,x5,x8. Instead of doing X = x2*0.707 + x5* 0.707 + x8* 0.707. FFT does x = 0.707 * x2+x5+x8. Which saves computational time. Overall the time complexity of FFT is N log2N , is much faster than DFT. For example, for N = 65,536 and a CPU cycle time of 1 ns (1GHz), the DFT would take (65,5362)−9 = 4.29s, while the FFT would take (65536log265536)−9 = 0.001s.

In conclusion FFT is significantly faster and it is advised to sample data at a binary rate 2^n , 2,....256, 1024, 2048 ,etc.

2.2.4 Applications of Spectrum analysis

2.2.4.2 Deciding the sampling rate :

The sampling Nyquist theorem , the book doesn't go into details but the sampling frequency must be at least twice as the highest frequency of input signal. Also higher sampling rates in video camera FPS there is a tradeoff between quality and higher sampling rate. As a higher sampling rate requires more computational and processing power. A 24 FPS camera was found to be good enough for standard and pathological gait studies.

2.2.4.1 Analog to Digital converters:

The way the digital computer understands our continuous world. For example a simple voltage measurement signal that changes over time or an encoder that measures joint angles also changing over time. The computer simply doesn't understand this; it understands zeros and ones only digital. So a brilliant idea the ADC works in sampling the Analog signal at a very high rate (not identical to the original but enough for the digital computer to understand it. See the figure below. The sampling rate is defined above.

2.2.4.3 Deciding the record length :

Decided by the lowers frequency present in the signal. Doesn't go into detail why the later is concluded. But Record time T = 1/Fl.
An example of standing still the lowest frequency is 0.0073 Hz , so the record time is 137 seconds. Experiments show that for statistical reliability it should be T= 1 min or longer but for the pathological population this time might be compromised because they might not be able to stand still for this much time.

2.2.4.4 Analog and digital filtering of signals Noise and movement artifacts :

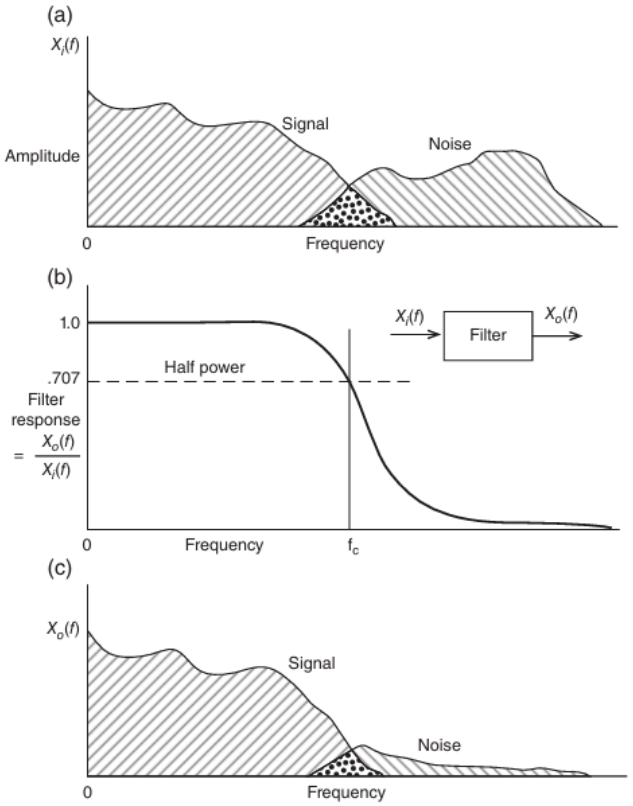


Figure 2.16 (a) Hypothetical frequency spectrum of a waveform consisting of a desired signal and unwanted higher frequency noise. (b) Response of low-pass filter $X_0(f)/X_i(f)$, introduced to attenuate the noise. (c) Spectrum of the output waveform, obtained by multiplying the amplitude of the input by the filter response at each frequency. Higher-frequency noise is severely attenuated, while the signal is passed with only minor distortion in transition region around f_c.

As in the figure there is some noise that we want to get rid of. So we introduce a low pass filter (I will not go into details of its calculations). Simply the low pass filter lets the low frequency signal pass and attenuates the higher frequency specifically at the cut off frequency Fc. For analog LP filters there is a tradeoff between reducing noise and not distorting the signal. As shown in the figure there is still some noise after applying the filter because if we apply the Fc with lower value it will attenuate some of our signal. And if we apply it with a higher value to keep our signal, we will let too much noise pass. Use of digital filters are sharper but they come with a higher computing cost.

2.2.4.5 Fourier Reconstitution of Original Signal.